GLOBAL ECONOMICS

10th grade

Term 2 – Lesson 3 MATERIAL: Decision Analysis with Bayes, Maximin, and Minimax Regret (Worked Solutions)

C9: Decision trees with staged filtering rules

1.

Problem description

A city logistics office must choose one policy among $B_1, B_2, B_3, B_4, B_5, B_6$. Payoffs (thousand USD) and probabilities are:

State	Probability	B_1	B_2	B_3	B_4	B_5	B_6
T_1	0.50	95	90	110	85	100	92
T_2	0.30	70	75	60	80	65	78
T_3	0.20	30	40	10	55	35	20

Ordered criterion results (already computed):

$$\text{EV ranking: } B_4(77.50) > B_5(76.50) > B_2(75.50) > B_3(75.00) > B_1(74.50) > B_6(73.40).$$

$$\text{Maximin ranking: } B_4(55) > B_2(40) > B_5(35) > B_1(30) > B_6(20) > B_3(10).$$

$$\text{Minimax Regret ranking: } B_2(20) = B_5(20) > B_1(25) = B_4(25) > B_6(35) > B_3(45).$$

Filtering rule: (1) keep the best 4 EV options; (2) from those, keep the best 2 Maximin options; (3) from those, choose the smallest Minimax Regret.

C9

Step 1. Reconfirm ordered lists.

- EV top 4: $\{B_4, B_5, B_2, B_3\}$.
- On this subset, Maximin values are $B_4(55),\ B_2(40),\ B_5(35),\ B_3(10)$, so top 2 are $\{B_4, B_2\}$.
- On $\{B_4, B_2\}$, Max regrets are $B_4(25),\ B_2(20)$.

Step 2. Decision tree for staged filtering.

Decision tree (filtering nodes and leaves)
Root decision node D_0: All alternatives $\{B_1, B_2, B_3, B_4, B_5, B_6\}$
$\rightarrow$ Node D_1 (EV filter): keep $\{B_4, B_5, B_2, B_3\}$
$\rightarrow$ Node D_2 (Maximin filter): keep $\{B_4, B_2\}$
$\rightarrow$ Node D_3 (Minimax Regret filter): choose smallest max regret
Leaves: B_2 (20), B_4 (25)
Final leaf selected: B_2

Step 3. Final decision. The decision tree selects B_2.

2.

Problem description

A regional distributor must choose one expansion plan among $X_1, X_2, X_3, X_4, X_5, X_6$. Payoffs (thousand USD) and probabilities are:

State	Probability	X_1	X_2	X_3	X_4	X_5	X_6
U_1	0.30	130	120	140	110	125	118
U_2	0.25	90	95	80	100	88	92
U_3	0.25	60	70	55	85	65	75
U_4	0.20	20	40	10	60	45	35

Ordered criterion results (already computed):

$$\text{EV ranking: } X_4(91.25) > X_2(85.25) > X_5(84.75) > X_6(84.15) > X_1(80.50) > X_3(77.75).$$

$$\text{Maximin ranking: } X_4(60) > X_5(45) > X_2(40) > X_6(35) > X_1(20) > X_3(10).$$

$$\text{Minimax Regret ranking: } X_2(20) = X_5(20) > X_6(25) > X_4(30) > X_1(40) > X_3(50).$$

Filtering rule: (1) keep the best 3 EV options; (2) from those, keep the best 2 Minimax Regret options; (3) from those, choose the best Maximin option.

C9

Step 1. Reconfirm ordered lists and filtering sets.

- EV top 3: $\{X_4, X_2, X_5\}$.
- In this subset, Minimax Regret values are $X_2(20), X_5(20), X_4(30)$, so keep $\{X_2, X_5\}$.
- In $\{X_2, X_5\}$, Maximin values are $X_2(40), X_5(45)$.

Step 2. Decision tree for staged filtering.

Decision tree (filtering nodes and leaves)
Root decision node D_0: $\{X_1, X_2, X_3, X_4, X_5, X_6\}$
$\rightarrow$ Node D_1 (EV filter): keep $\{X_4, X_2, X_5\}$
$\rightarrow$ Node D_2 (Minimax Regret filter): keep $\{X_2, X_5\}$
$\rightarrow$ Node D_3 (Maximin choice)
Leaves: X_2 (min = 40), X_5 (min = 45)
Final leaf selected: X_5

Step 3. Final decision. The decision tree selects X_5.

3.

Problem description

A trade finance agency chooses one guarantee policy among $G_1, G_2, G_3, G_4, G_5, G_6$. States and probabilities are shown below. Payoffs are in thousand USD. Policies G_4 and G_5 balance upside and downside because they diversify by sector, while G_3 is highly cyclical.

State	Probability	G_1	G_2	G_3	G_4	G_5	G_6
M_1	0.45	100	94	114	90	104	96
M_2	0.35	72	78	64	84	70	80
M_3	0.20	32	44	12	56	38	26

Ordered criterion results (already computed):

$$\text{EV ranking: } G_4(83.90) > G_5(81.70) > G_2(81.20) > G_3(79.70) > G_1(78.10) > G_6(75.40).$$

$$\text{Maximin ranking: } G_4(56) > G_2(44) > G_5(38) > G_1(32) > G_6(26) > G_3(12).$$

$$\text{Minimax Regret ranking: } G_2(20) = G_5(20) > G_4(24) > G_1(24) > G_6(34) > G_3(44).$$

Filtering rule: (1) keep best 4 EV options; (2) from those, keep best 2 Maximin options; (3) from those, choose smallest Minimax Regret.

C9

Step 1. Reconfirm ordered lists.

- EV top 4: $\{G_4, G_5, G_2, G_3\}$.
- On this subset, Maximin values are $G_4(56), G_2(44), G_5(38), G_3(12)$, so keep $\{G_4, G_2\}$.
- On $\{G_4, G_2\}$, max regrets are $G_4(24), G_2(20)$.

Step 2. Decision tree for staged filtering.

Decision tree (filtering nodes and leaves)
Root decision node D_0: all alternatives $\{G_1, G_2, G_3, G_4, G_5, G_6\}$
$\rightarrow$ Node D_1 (EV filter): keep $\{G_4, G_5, G_2, G_3\}$
$\rightarrow$ Node D_2 (Maximin filter): keep $\{G_4, G_2\}$
$\rightarrow$ Node D_3 (Minimax Regret filter): choose smallest max regret
Leaves: G_2 (20), G_4 (24)
Final leaf selected: G_2

Step 3. Final decision. The decision tree selects G_2.

4.

Problem description

An insurance broker considers six portfolio mixes Y_1 to Y_6 under four climate-risk states with probabilities W_1:0.30, W_2:0.25, W_3:0.25, W_4:0.20. Payoffs are in thousand USD. Y_4 is conservative and protects downside in severe-weather years, while Y_3 has the highest upside but a weak stress outcome.

State	Probability	Y_1	Y_2	Y_3	Y_4	Y_5	Y_6
W_1	0.30	136	126	146	116	130	121
W_2	0.25	94	98	84	103	90	95
W_3	0.25	66	74	58	88	69	77
W_4	0.20	28	46	14	64	48	38

Ordered criterion results (already computed):

$$\text{EV ranking: } Y_4(95.25) > Y_2(89.20) > Y_5(88.70) > Y_6(87.95) > Y_1(84.90) > Y_3(82.20).$$

$$\text{Maximin ranking: } Y_4(64) > Y_5(48) > Y_2(46) > Y_6(38) > Y_1(28) > Y_3(14).$$

$$\text{Minimax Regret ranking: } Y_2(20) = Y_5(20) > Y_6(26) > Y_4(30) > Y_1(40) > Y_3(50).$$

Filtering rule: (1) keep the best 3 EV options; (2) from those, keep the best 2 Minimax Regret options; (3) from those, choose the best Maximin option.

C9

Step 1. Reconfirm ordered lists and filtering sets.

- EV top 3: $\{Y_4, Y_2, Y_5\}$.
- In this subset, Minimax Regret values are $Y_2(20), Y_5(20), Y_4(30)$, so keep $\{Y_2, Y_5\}$.
- In $\{Y_2, Y_5\}$, Maximin values are $Y_2(46), Y_5(48)$.

Step 2. Decision tree for staged filtering.

Decision tree (filtering nodes and leaves)
Root decision node D_0: $\{Y_1, Y_2, Y_3, Y_4, Y_5, Y_6\}$
$\rightarrow$ Node D_1 (EV filter): keep $\{Y_4, Y_2, Y_5\}$
$\rightarrow$ Node D_2 (Minimax Regret filter): keep $\{Y_2, Y_5\}$
$\rightarrow$ Node D_3 (Maximin choice)
Leaves: Y_2 (min = 46), Y_5 (min = 48)
Final leaf selected: Y_5

Step 3. Final decision. The decision tree selects Y_5.